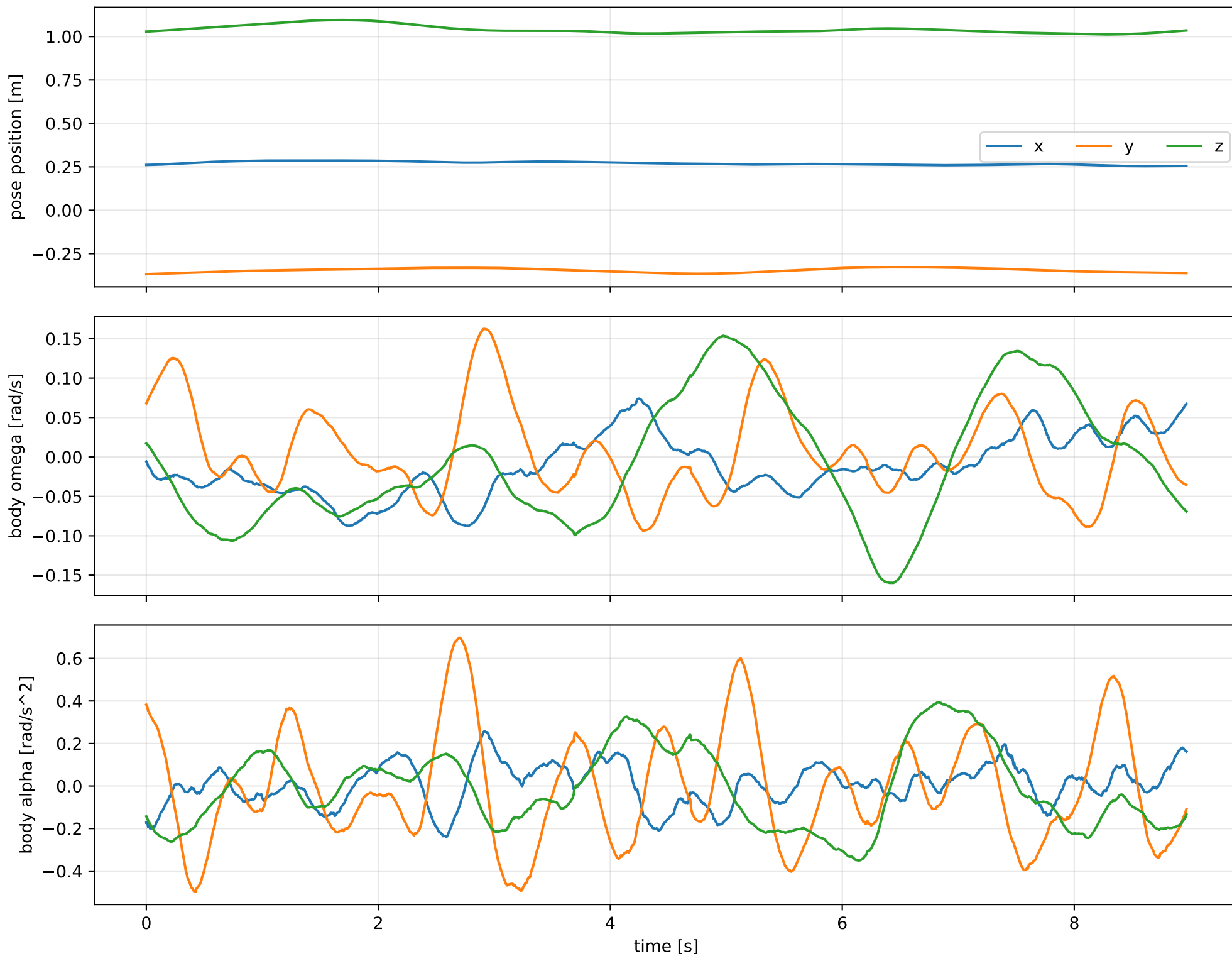
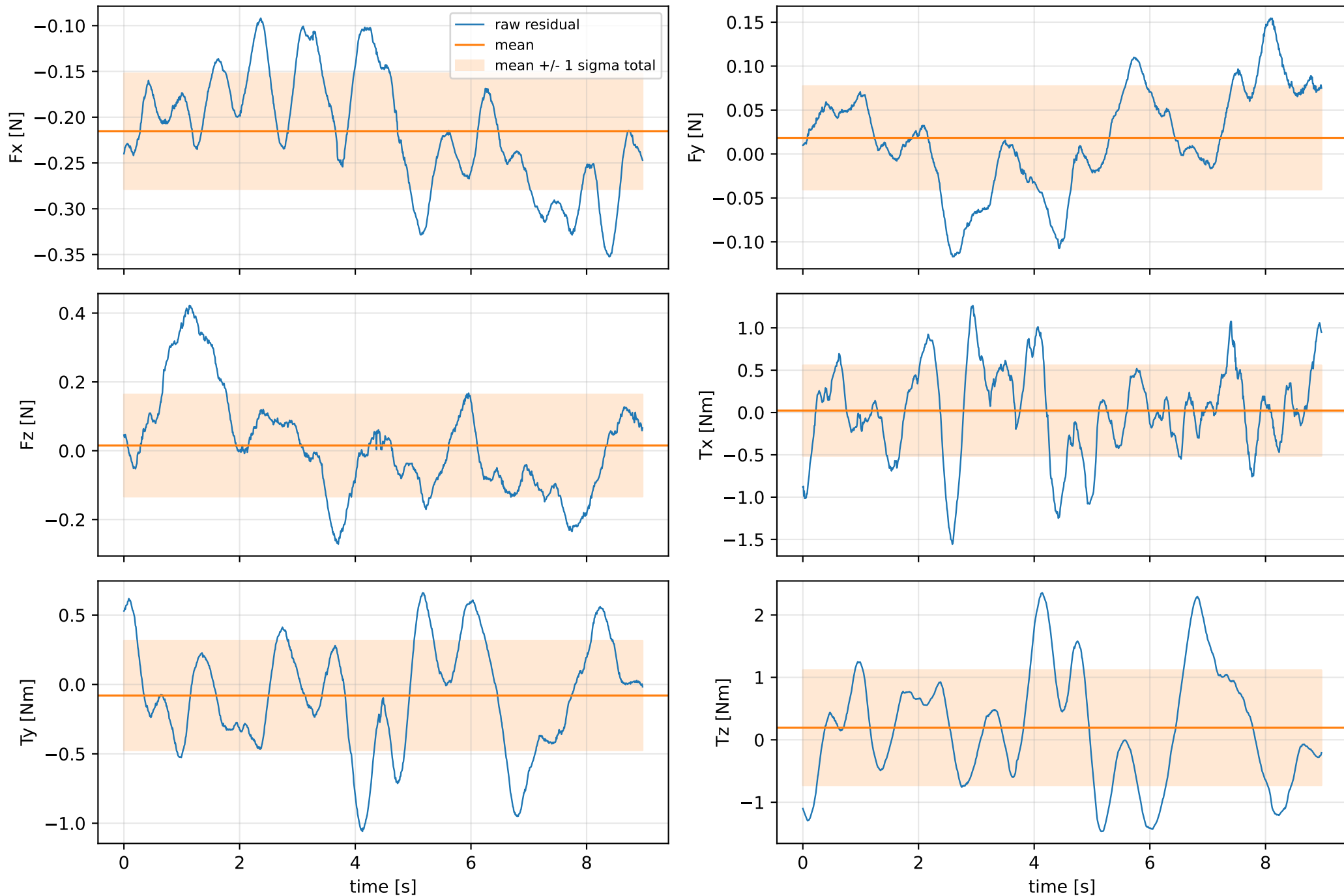


force_over_mass_rotor_3_nominal: SG trajectory and derived motion



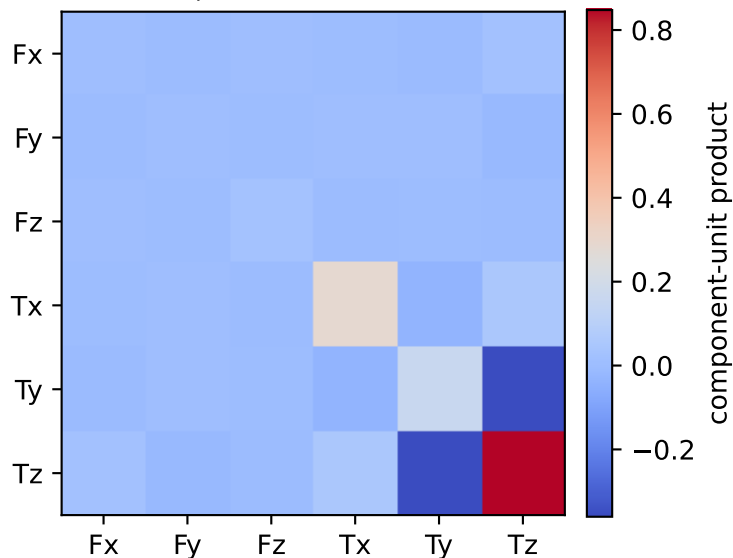
force_over_mass_rotor_3_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



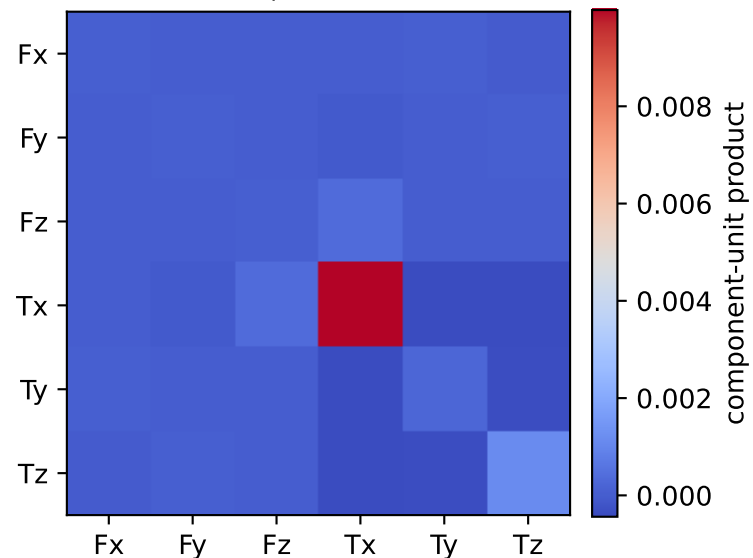
vector RMS about zero: force 0.27642 N, torque 1.1541 Nm; component std about mean: force [0.0632 0.0588 0.1484], torque [0.5344 0.3946 0.9207]

force_over_mass_rotor_3_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [4.473e-04 1.833e-03 4.344e-03 2.248e-02 2.724e-01 1.005e+00]

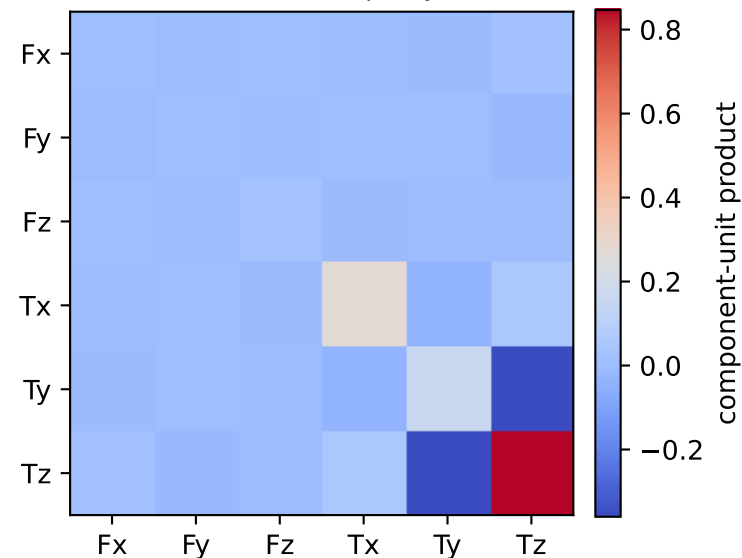
empirical total covariance



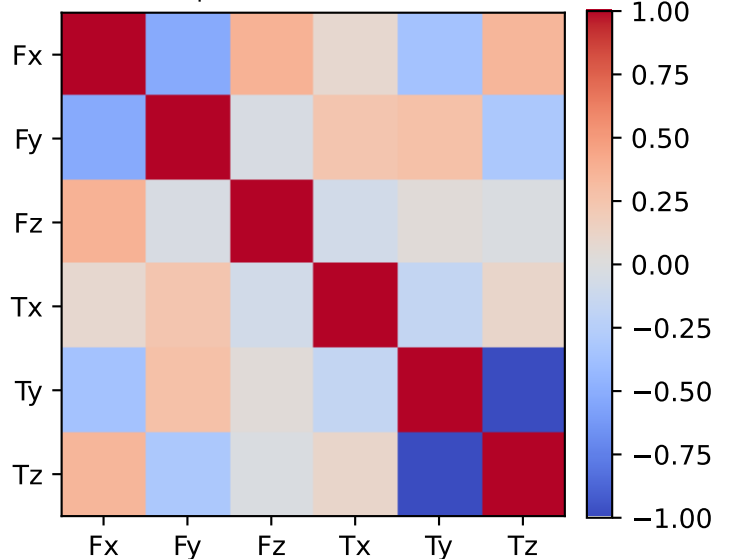
mean full-SG prediction covariance



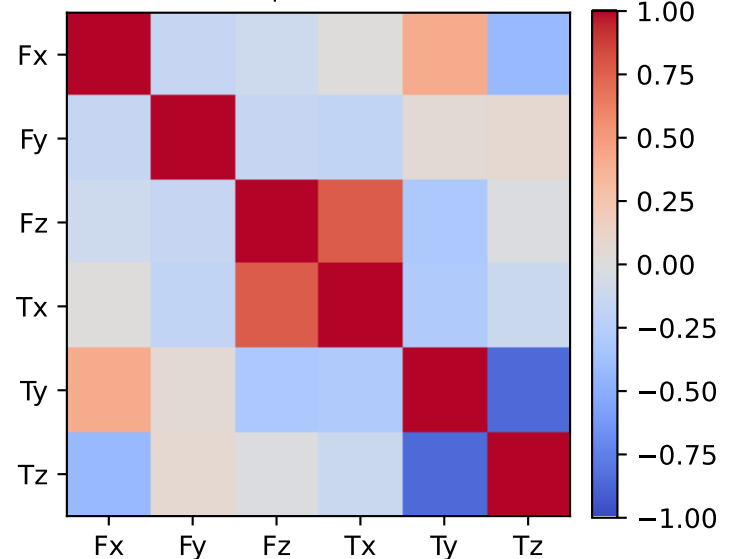
PSD excess model discrepancy covariance



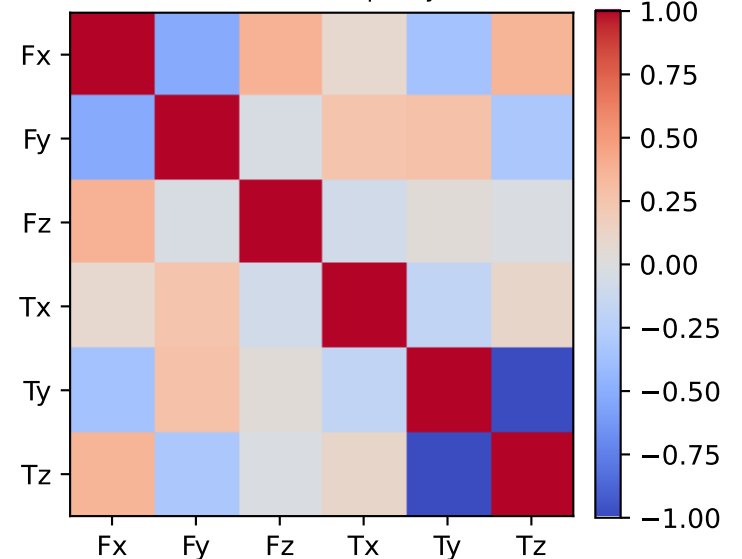
empirical total correlation



mean full-SG prediction correlation

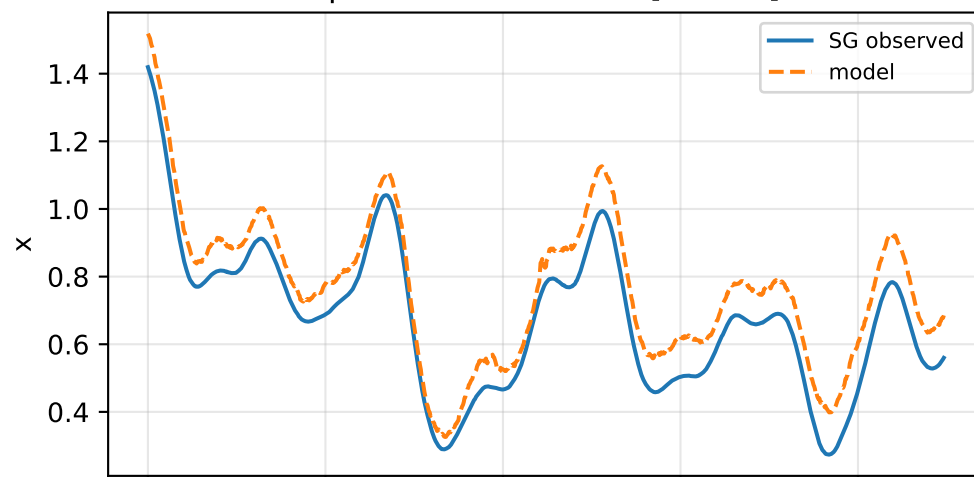


PSD excess model discrepancy correlation

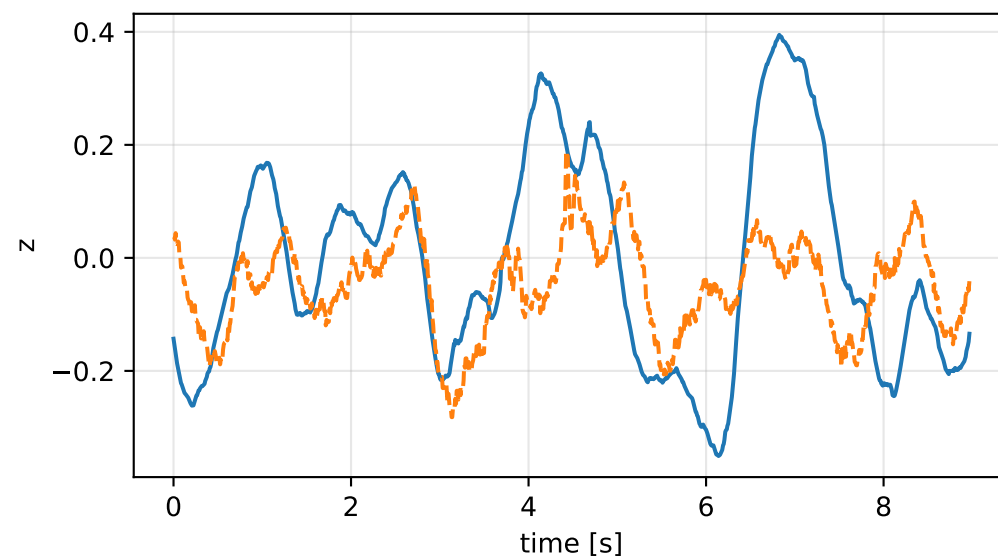
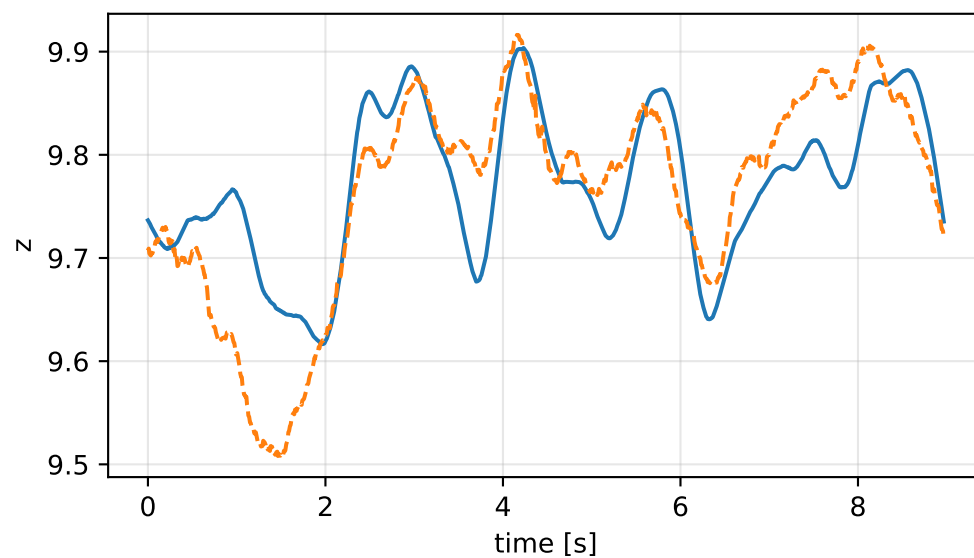
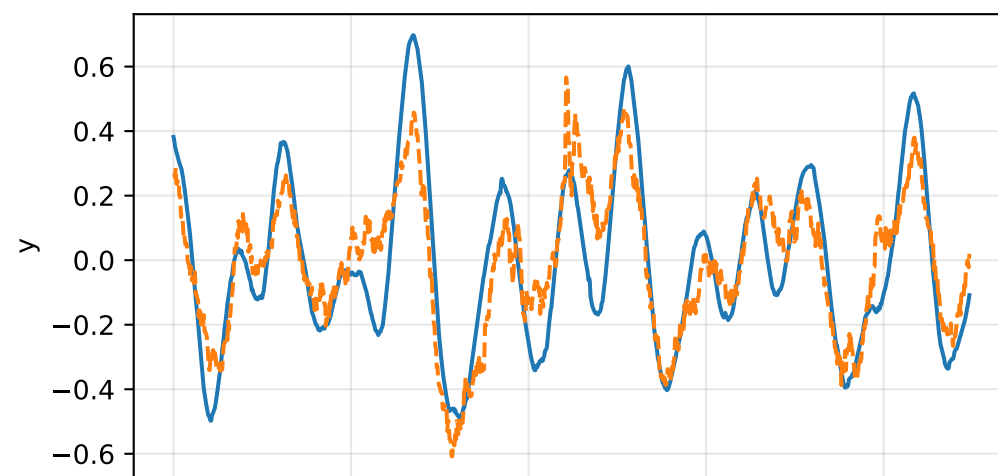
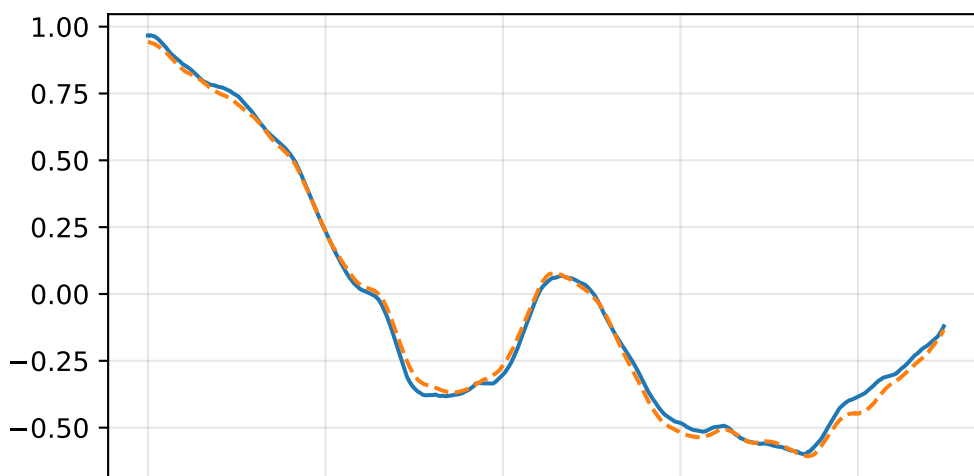
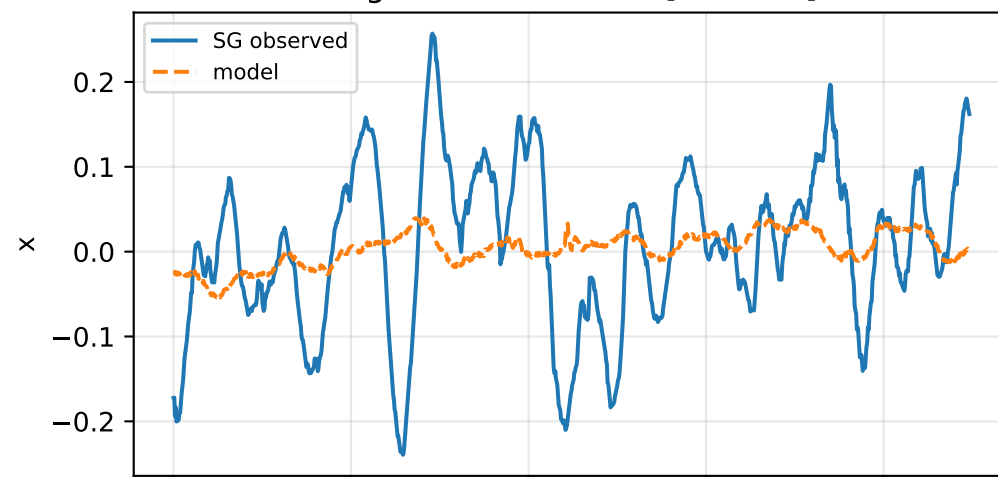


force_over_mass_rotor_3_nominal: acceleration residual objective

specific acceleration [m/s^2]

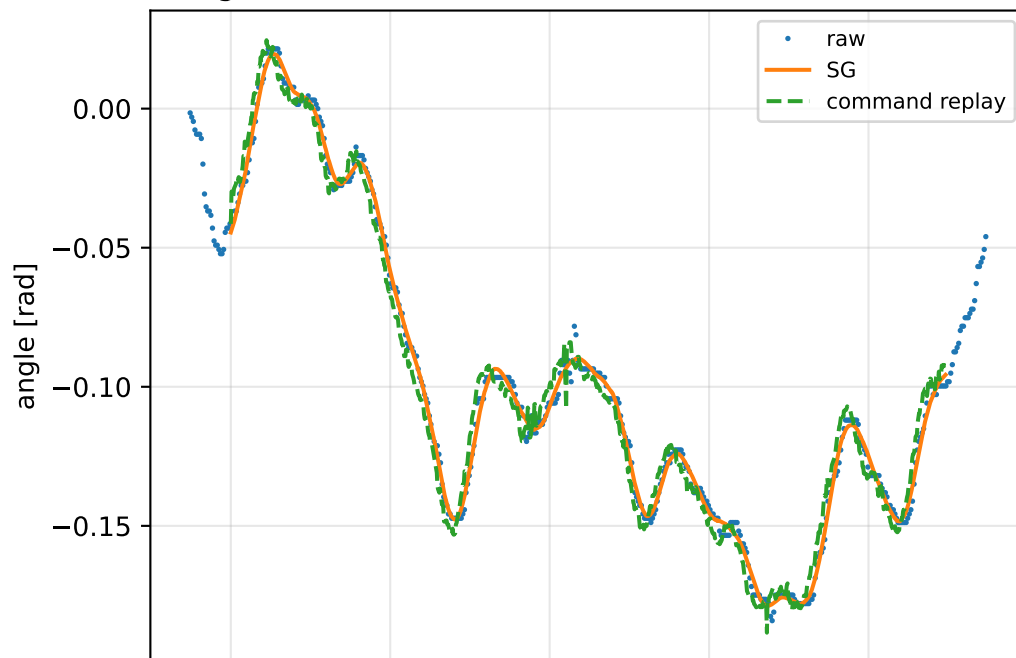


angular acceleration [rad/s^2]

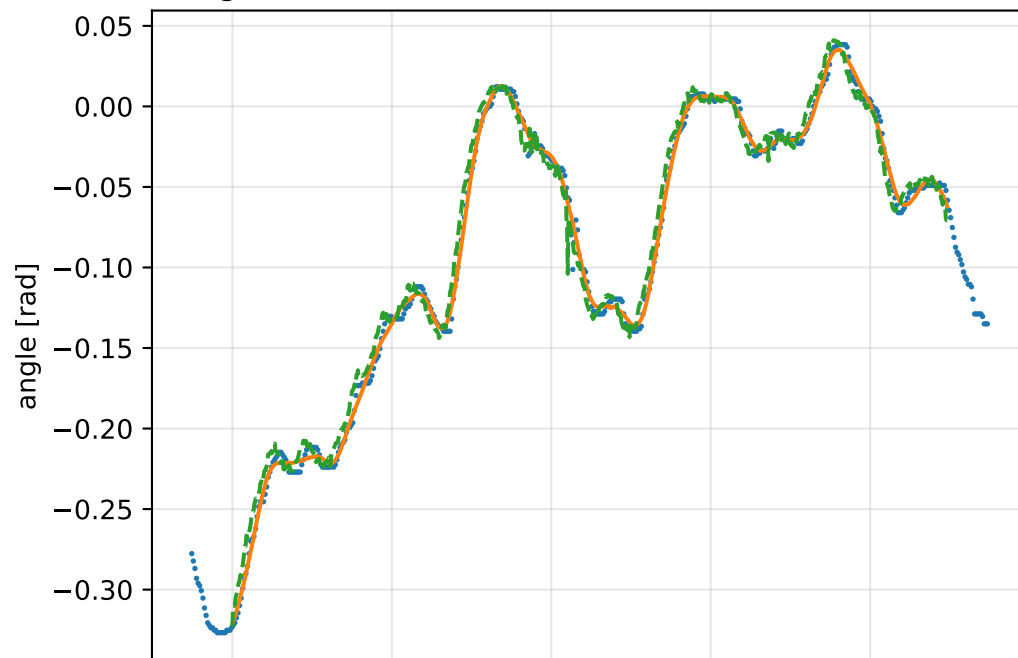


force_over_mass_rotor_3_nominal: gimbal measurement audit (objective=measured_sg)

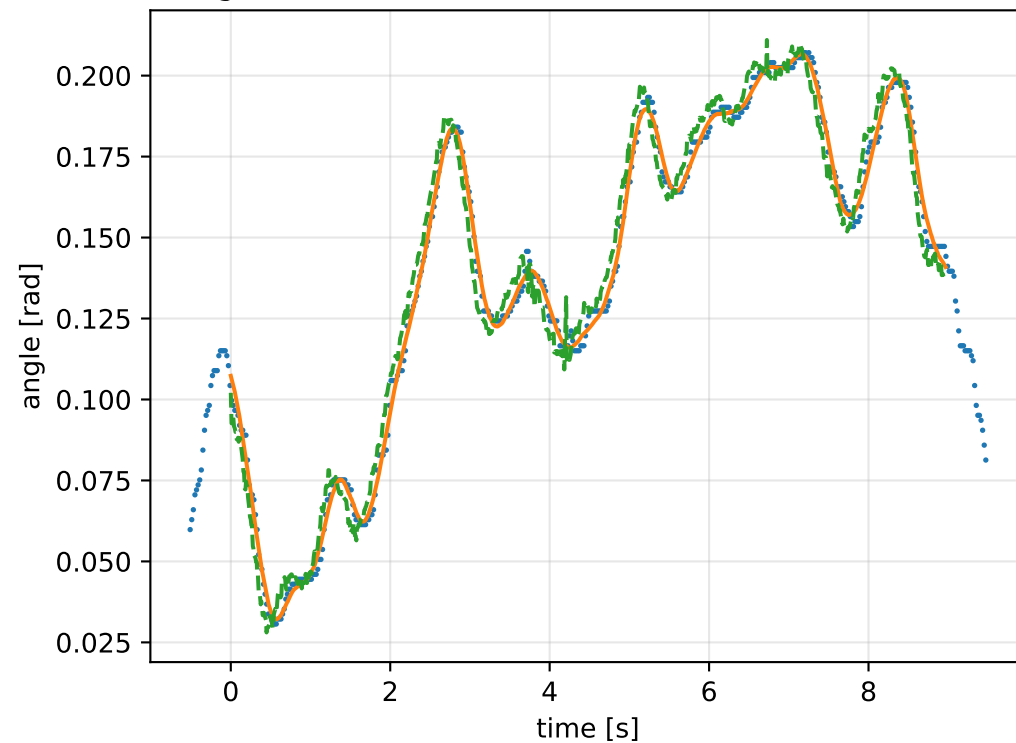
gimbal 1: RMSE=0.006345, max=0.01494 rad



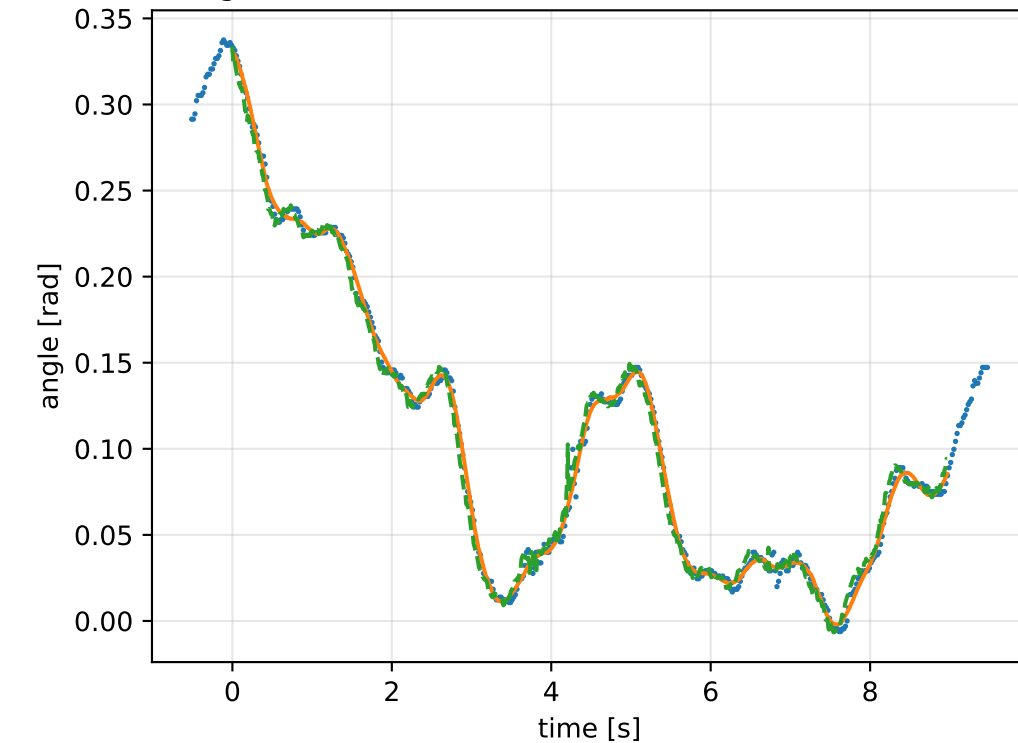
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

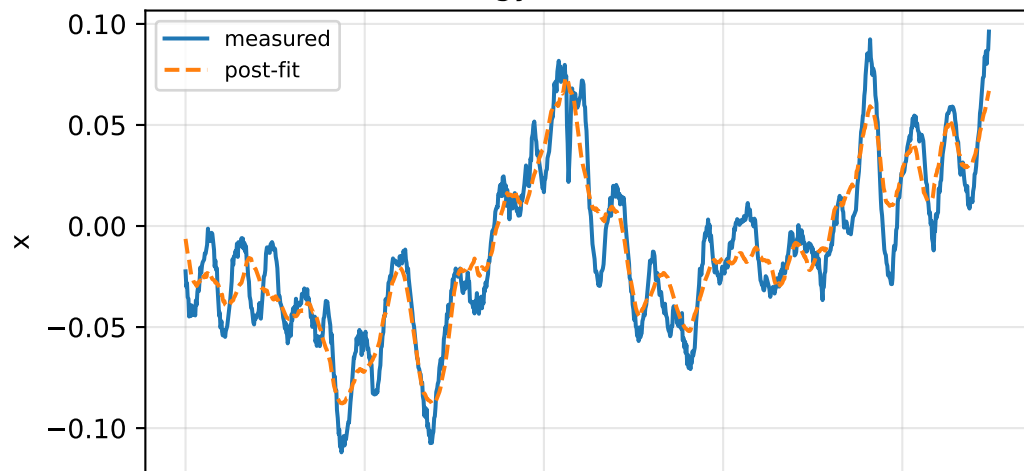


gimbal 4: RMSE=0.006321, max=0.03642 rad

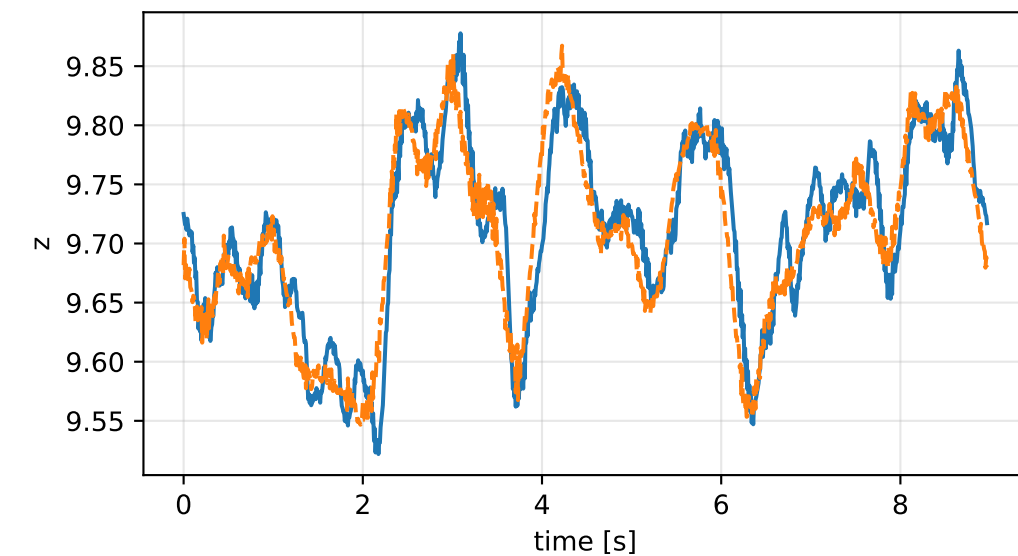
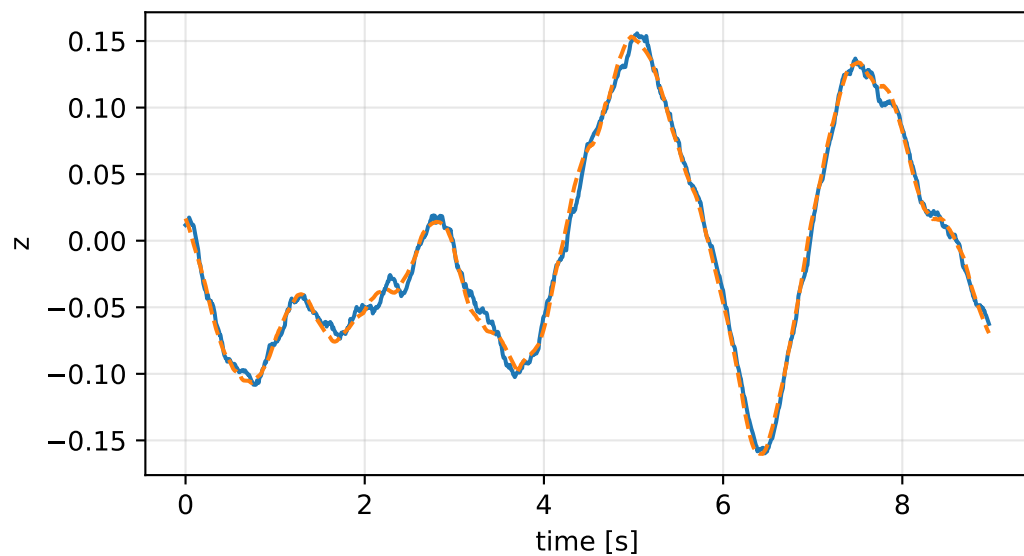
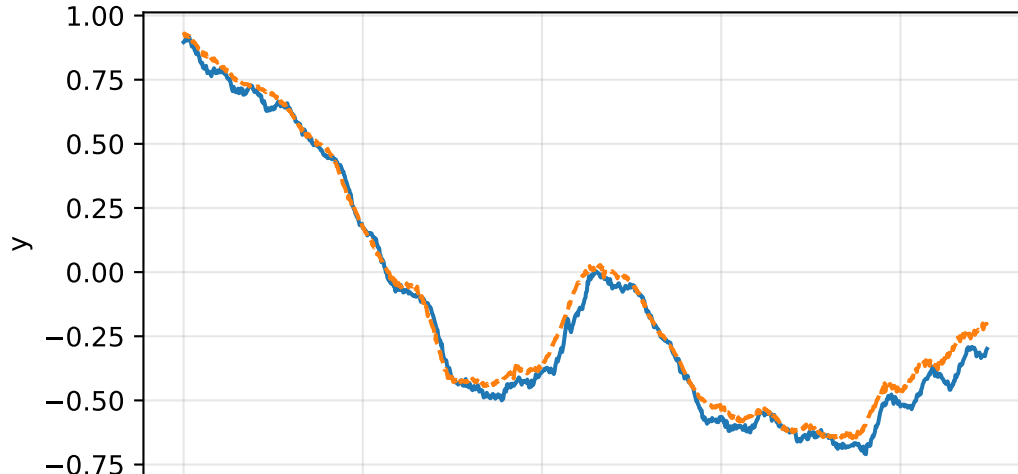
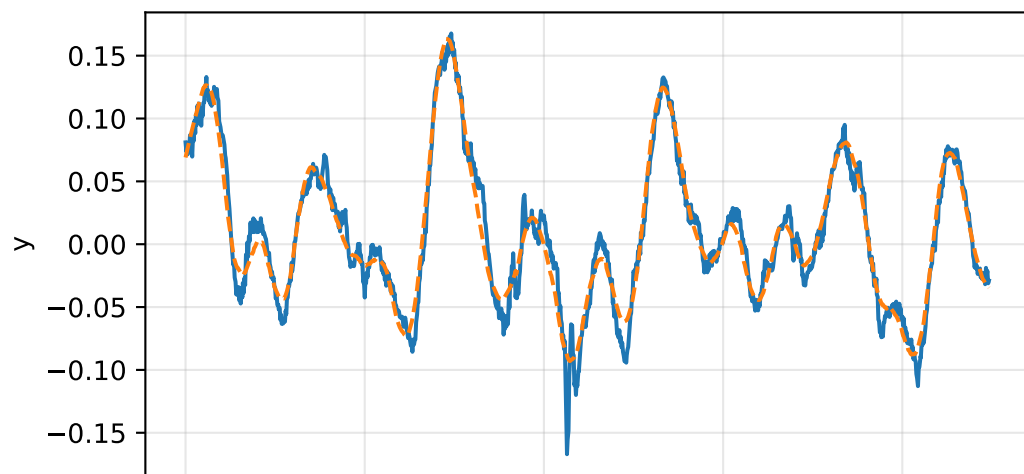
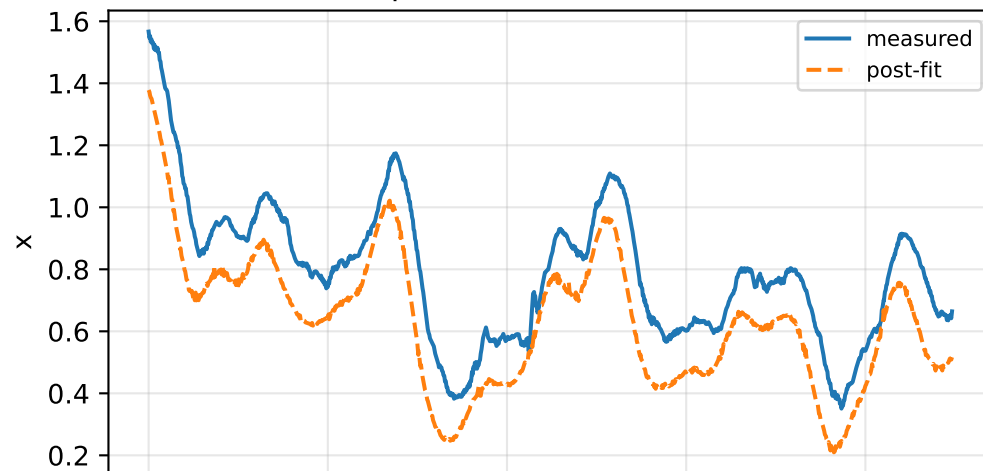


force_over_mass_rotor_3_nominal: IMU comparison (diagnostic only)

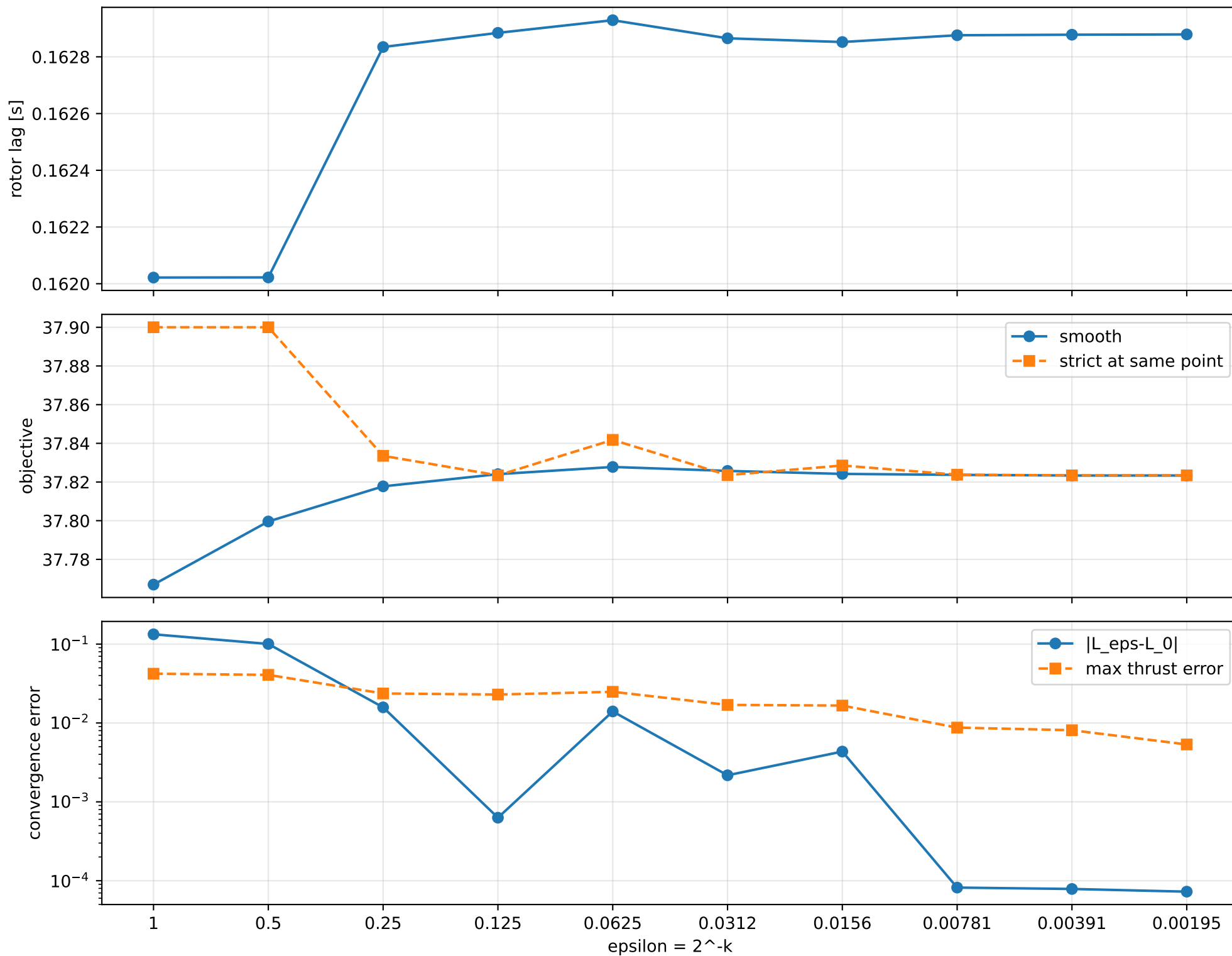
gyro [rad/s]



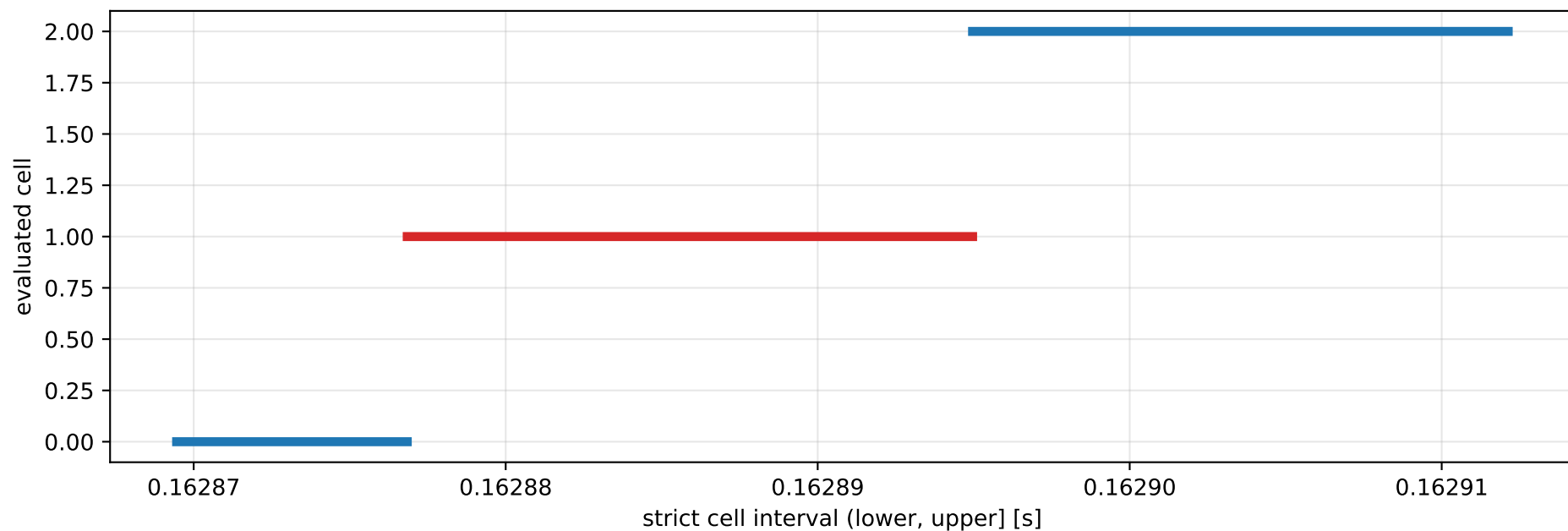
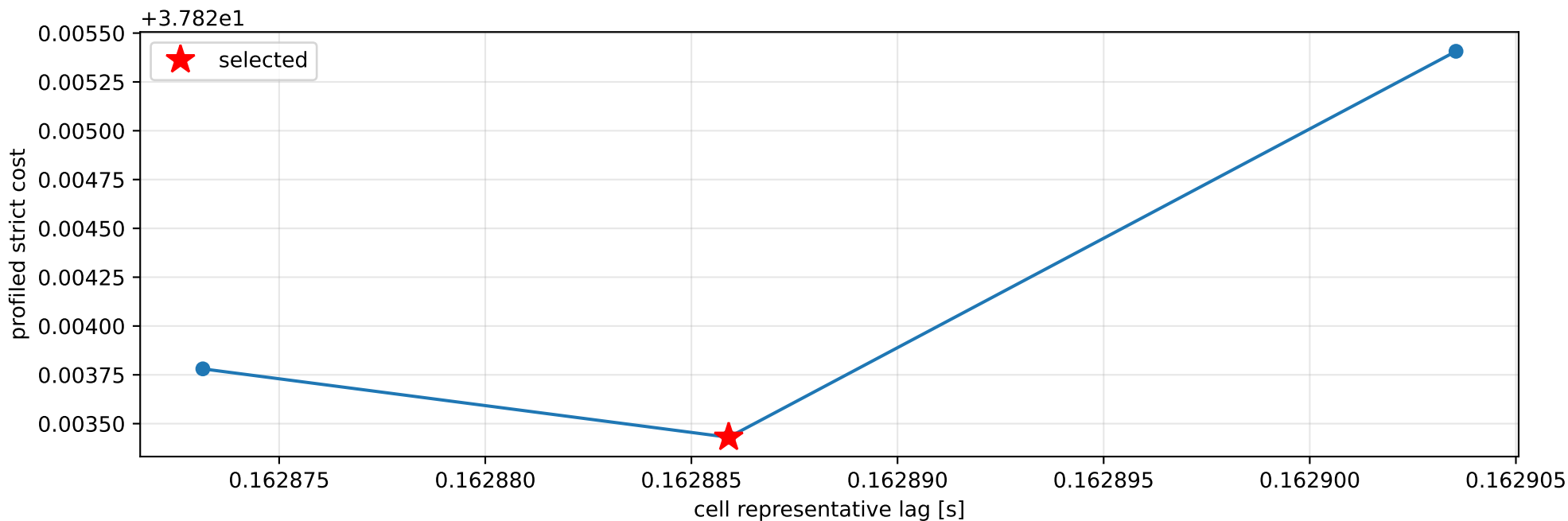
specific force [m/s²]



force_over_mass_rotor_3_nominal: smooth-to-strict continuation



force_over_mass_rotor_3_nominal: exact strict-ZOH lag cells



selected (0.162876844, 0.162894964] s; final smooth 0.162879057 s; data boundary=False

force_over_mass_rotor_3_nominal: parameter estimate

Case: force_over_mass_rotor_3_nominal
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 1.64907567

inertia [kg m²]:

```
[[ 3.98417694 -0.2519993 -0.00871383]
 [-0.2519993  0.7705002 -1.36455114]
 [-0.00871383 -1.36455114  3.42618934]]
```

CoG body [m]: [-0.00616527 0.0565721 -0.0613031]

force effectiveness: [0.40732727 0.46037866 0.70125759 0.704029]

scale-free J/m [m²]:

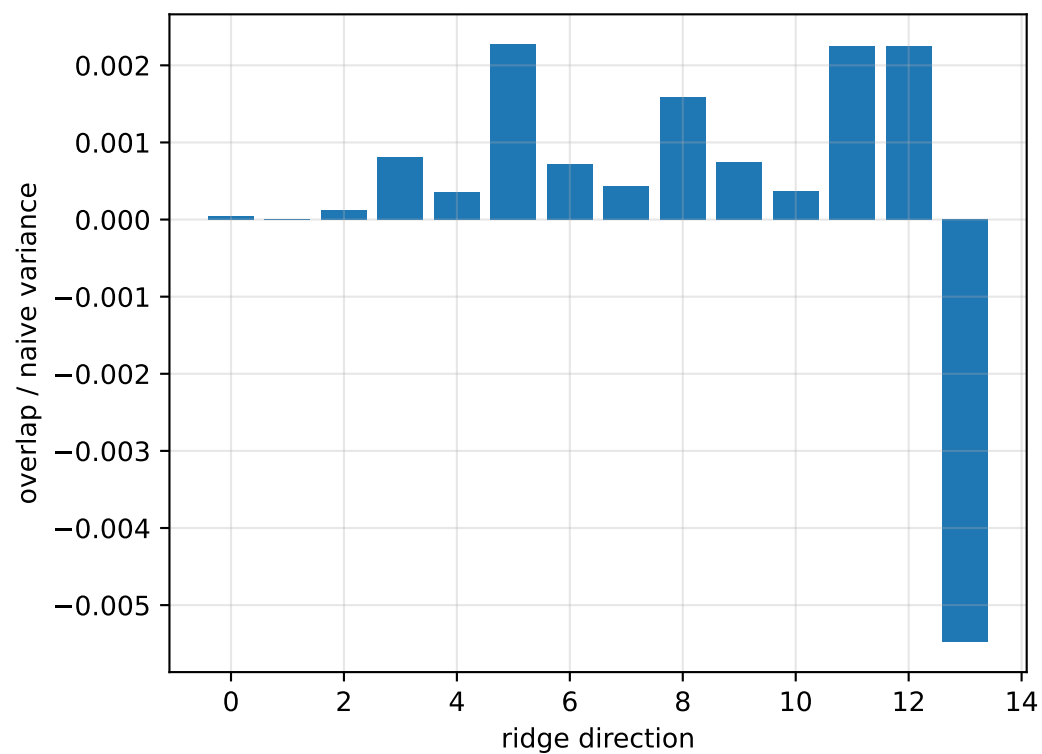
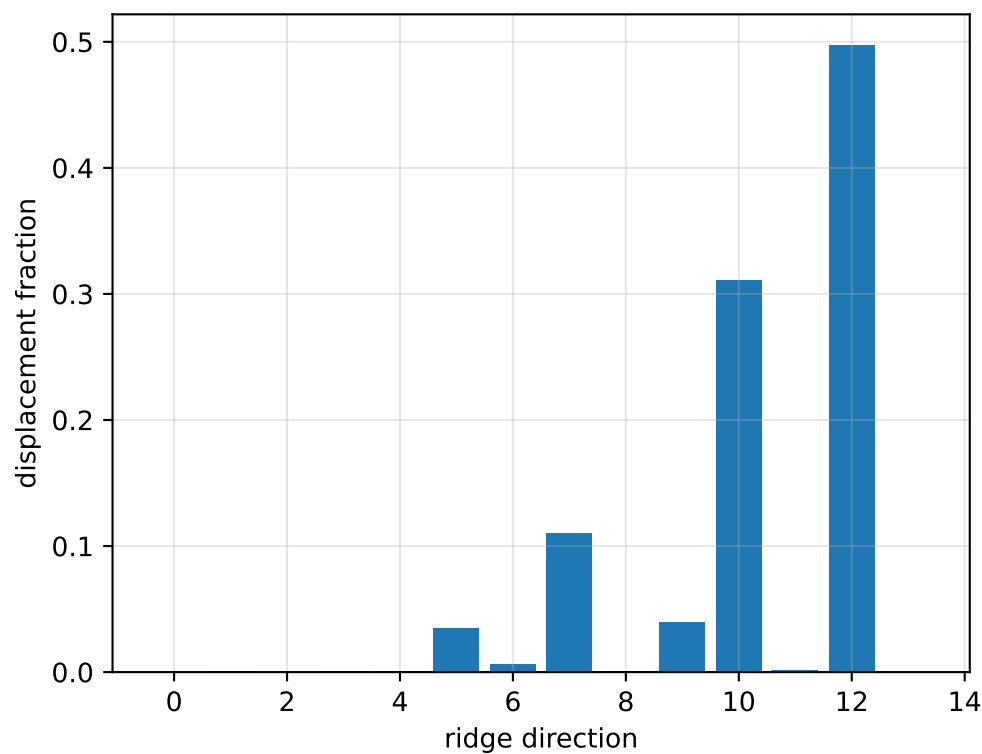
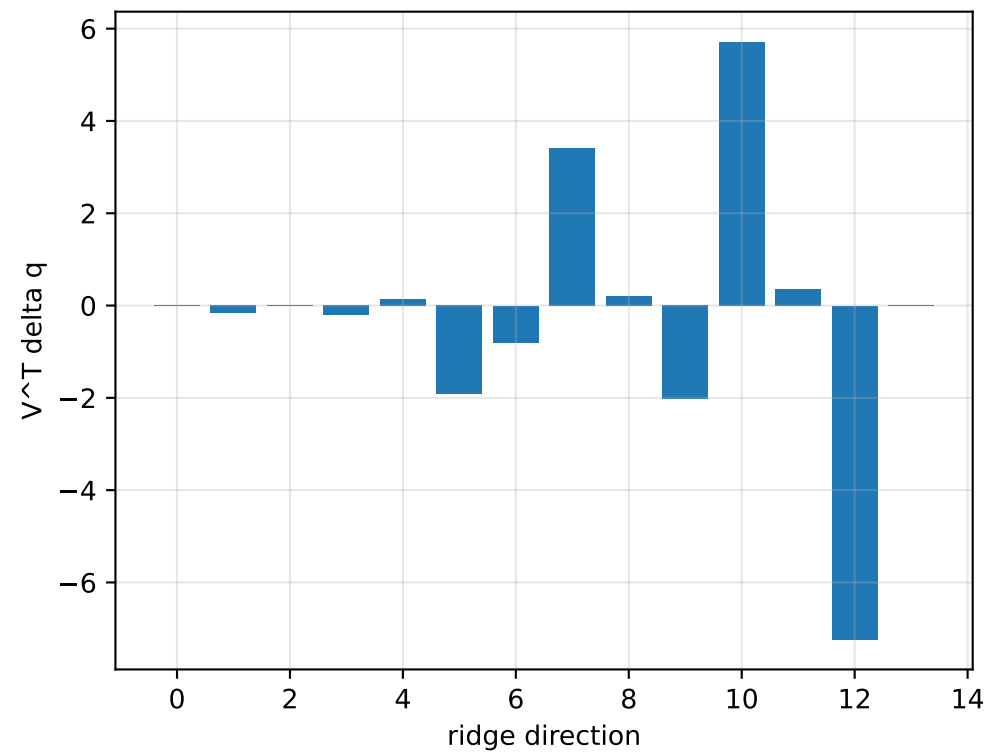
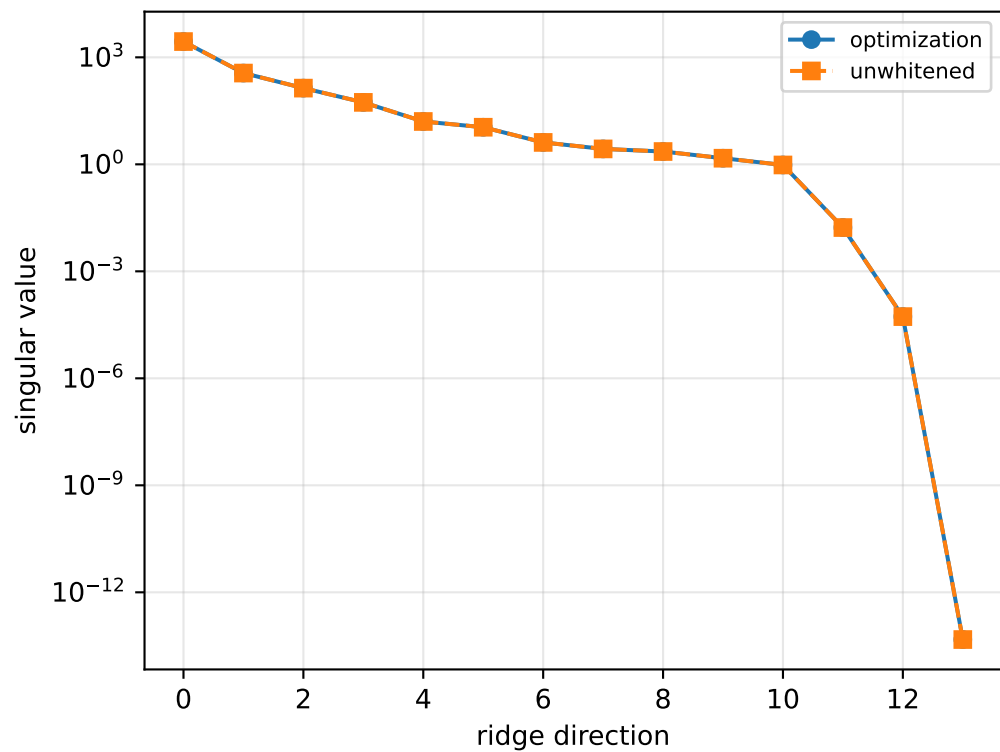
```
[[ 2.41600614 -0.15281245 -0.00528407]
 [-0.15281245  0.46723156 -0.82746424]
 [-0.00528407 -0.82746424  2.07764228]]
```

scale-free f/m: [0.24700338 0.27917376 0.42524282 0.42692341]

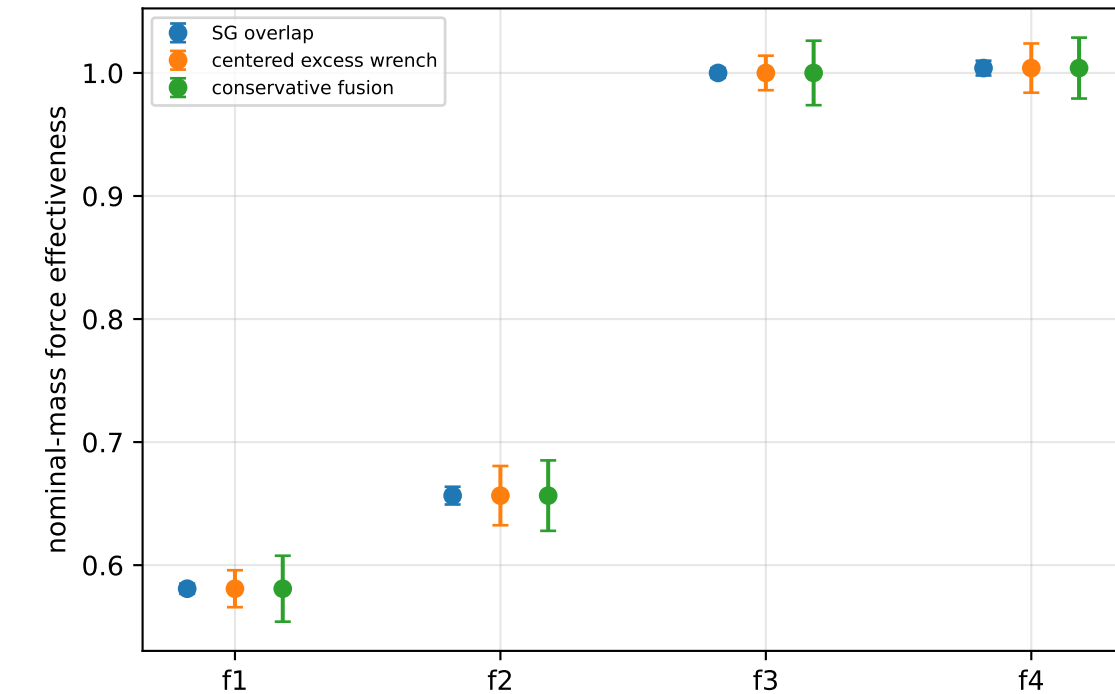
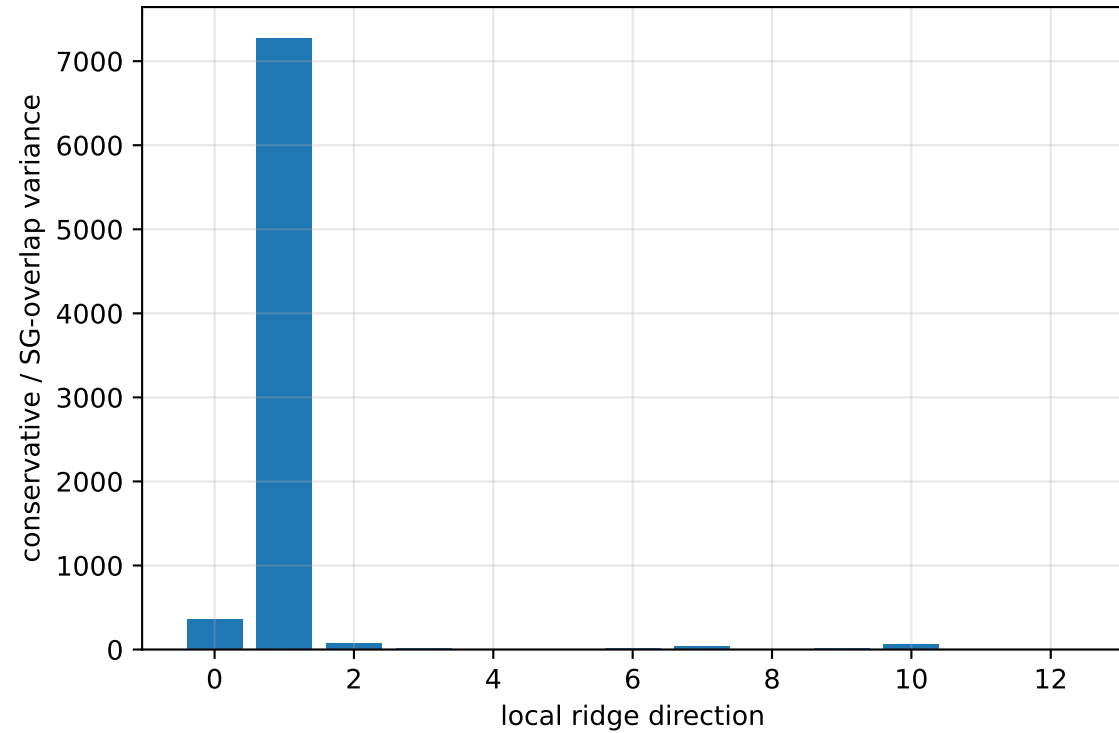
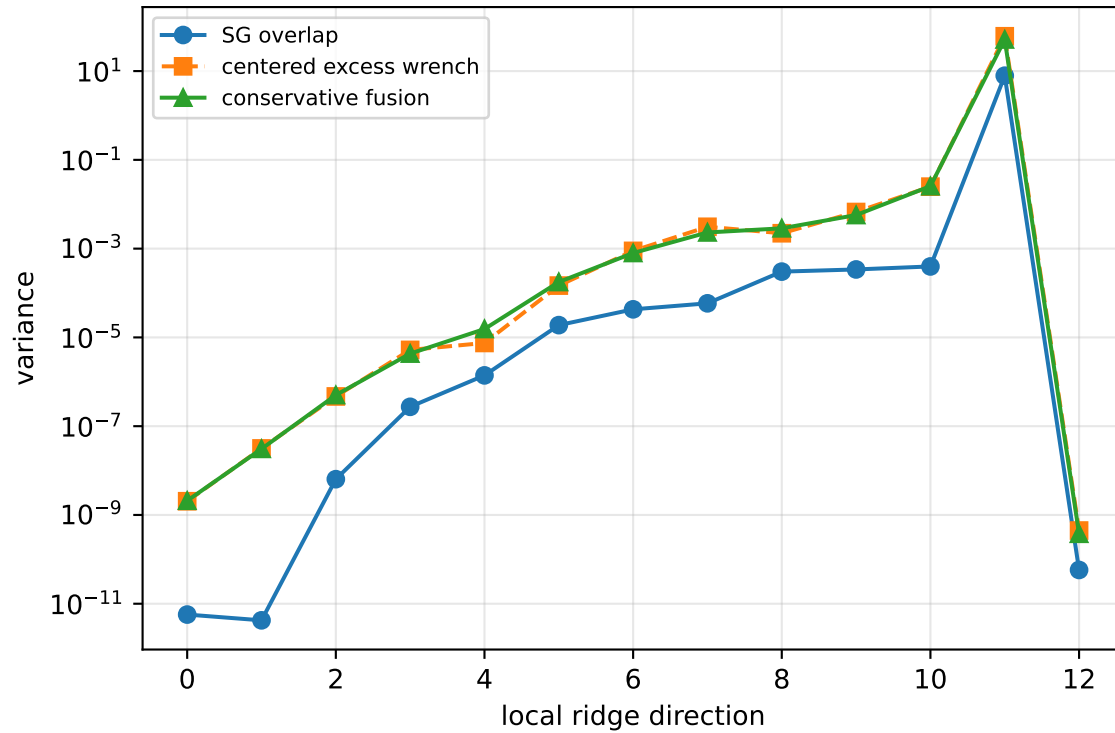
rotor lag cell: (0.162876844, 0.162894964] s

representative: 0.162885904 s

force_over_mass_rotor_3_nominal: ridge spectrum (rank 13, nullity 1, ||Jv||=7.82e-14)



force_over_mass_rotor_3_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 362.5$
wrench-to-acceleration recovery max error = $2.96\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

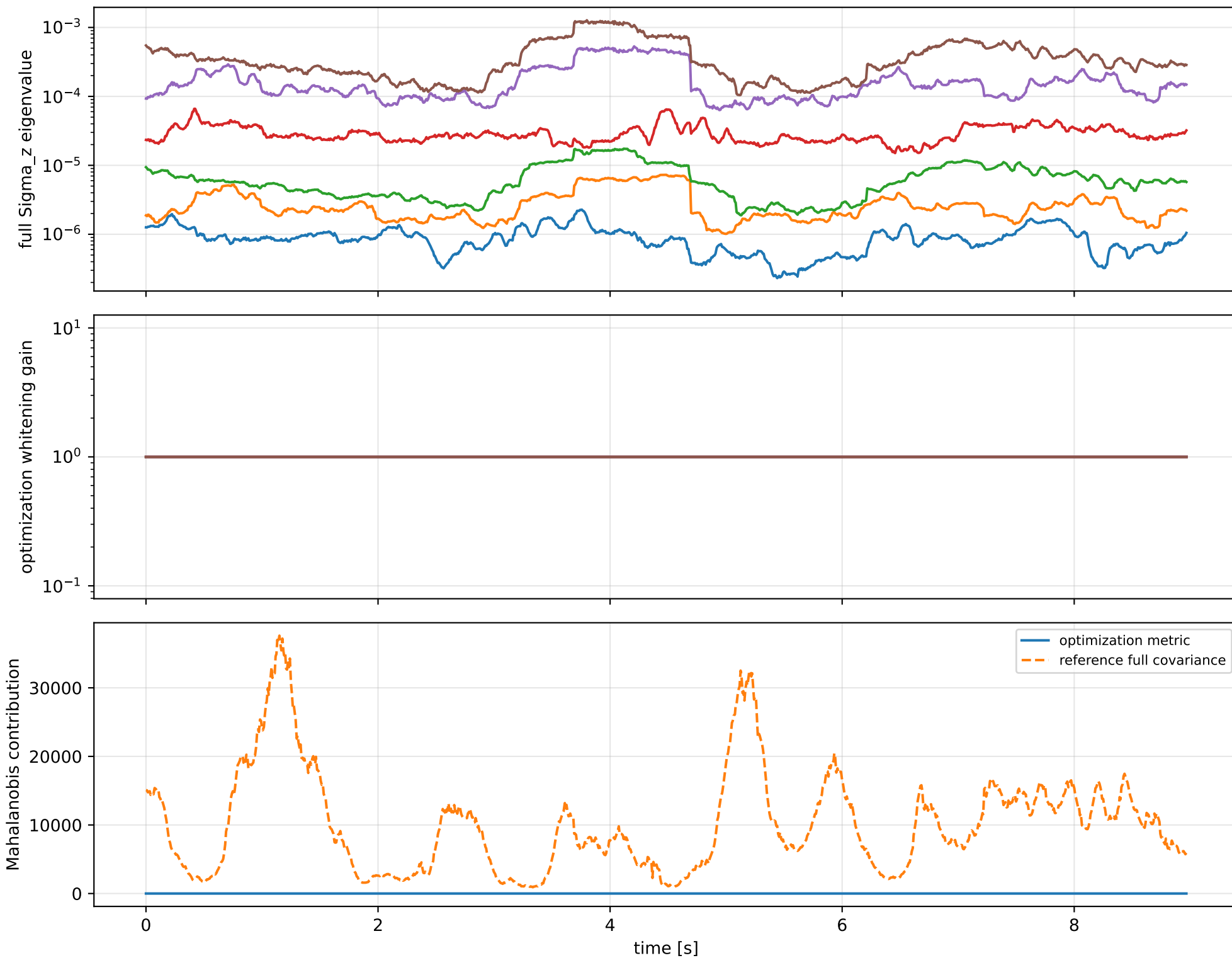
index 11: variance=51.42, ratio=6.531
 $\log_{\text{mass}} = -0.024$; second_moment_diag_1=-0.512
second_moment_diag_2=+0.598; second_moment_diag_3=+0.0344
second_moment_offdiag_12=-0.566; second_moment_offdiag_13=+0.128
second_moment_offdiag_23=-0.198; cog_x=+0.000119
cog_y=-8.21e-05; cog_z=+0.000101
 $\log_{\text{force_eff_1}} = -0.0243$; $\log_{\text{force_eff_2}} = -0.0229$
 $\log_{\text{force_eff_3}} = -0.0239$; $\log_{\text{force_eff_4}} = -0.0247$

index 10: variance=0.02552, ratio=64.29
 $\log_{\text{mass}} = -0.0864$; second_moment_diag_1=-0.0324
second_moment_diag_2=-0.237; second_moment_diag_3=+0.838
second_moment_offdiag_12=-0.0607; second_moment_offdiag_13=+0.311
second_moment_offdiag_23=-0.0439; cog_x=-0.00117
cog_y=+0.0266; cog_z=-0.0212
 $\log_{\text{force_eff_1}} = -0.257$; $\log_{\text{force_eff_2}} = -0.25$
 $\log_{\text{force_eff_3}} = +0.0077$; $\log_{\text{force_eff_4}} = +0.0169$

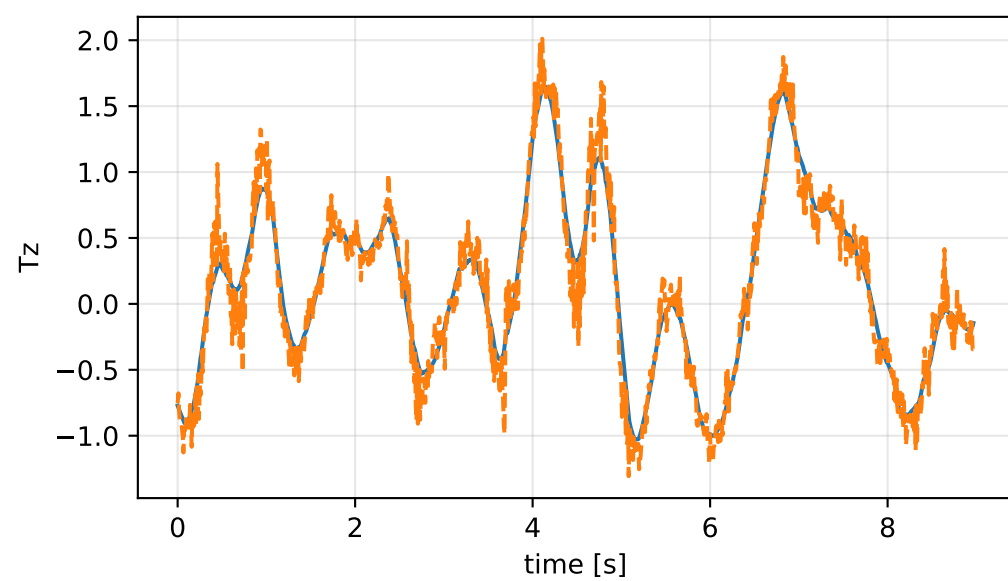
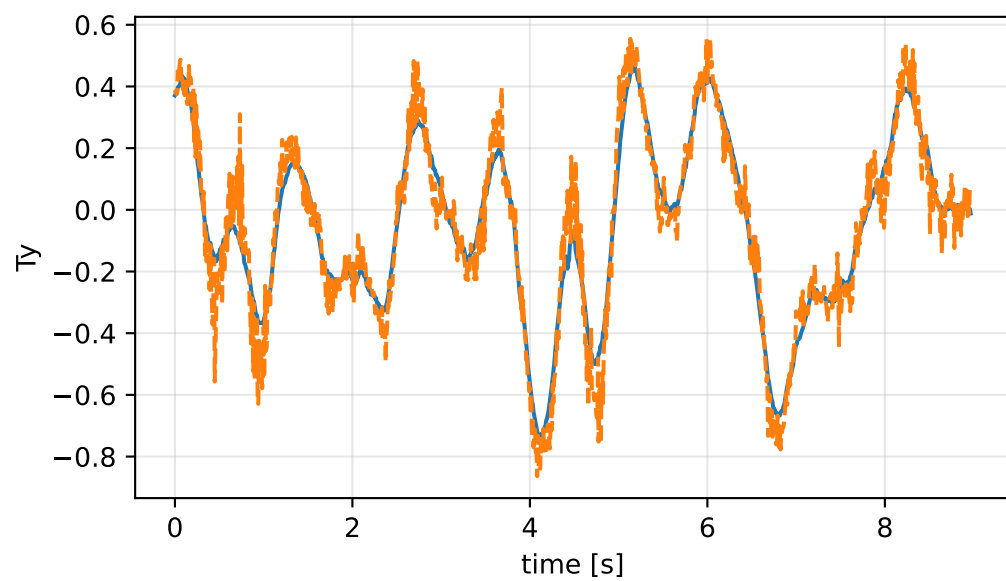
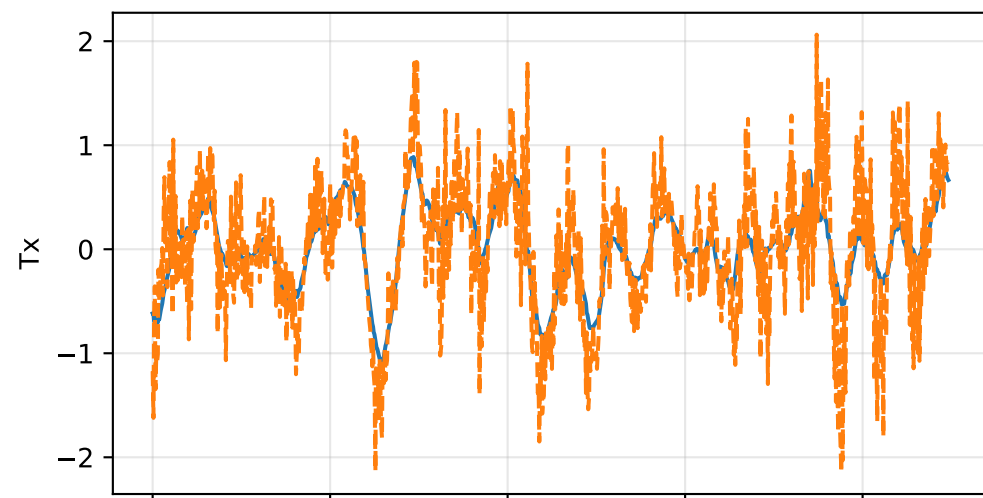
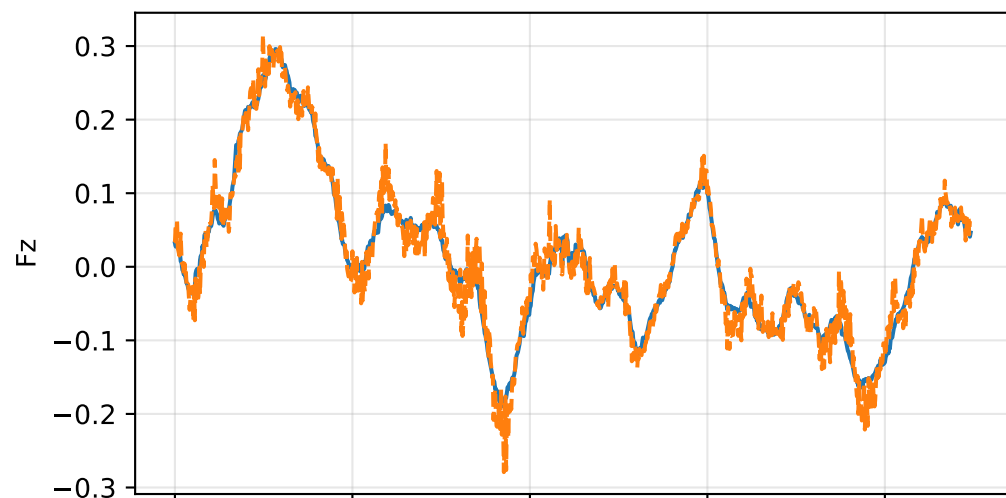
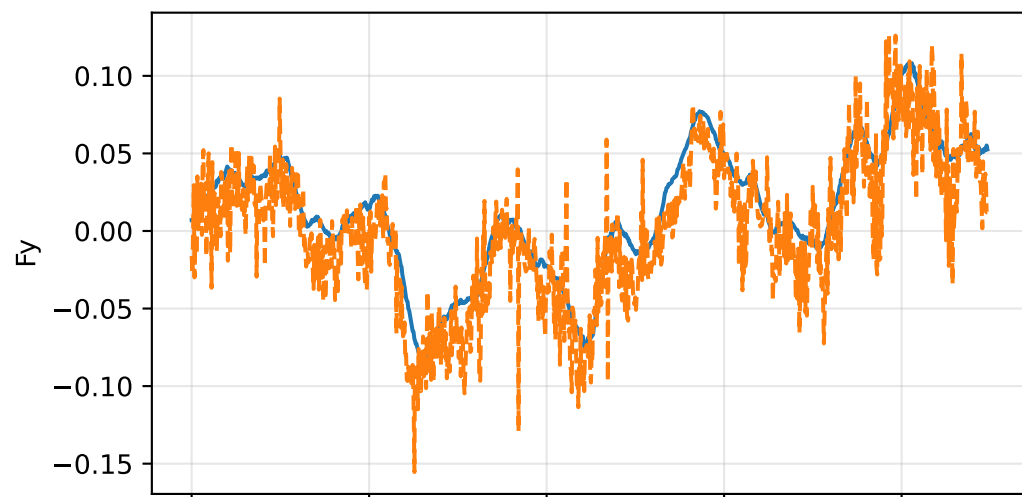
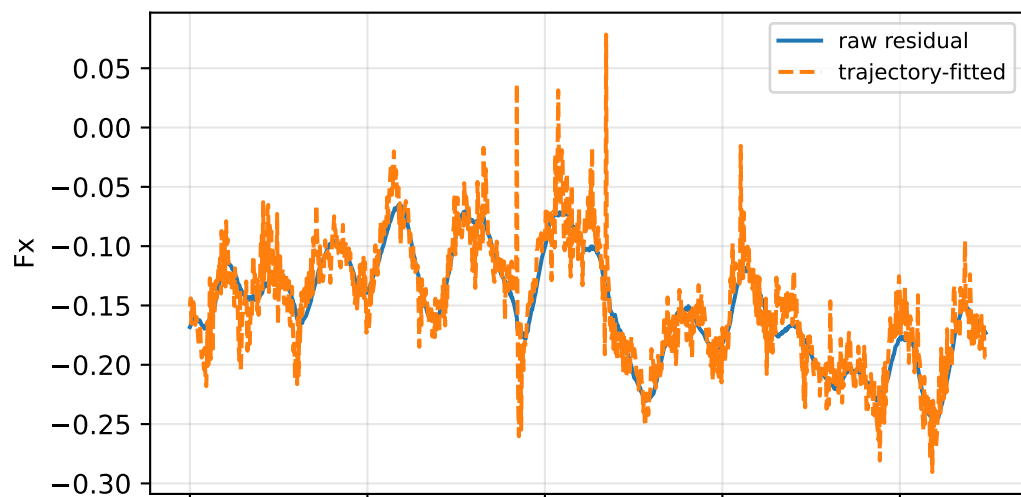
index 9: variance=0.005708, ratio=16.81
 $\log_{\text{mass}} = -0.00297$; second_moment_diag_1=+0.00106
second_moment_diag_2=-0.229; second_moment_diag_3=-0.0491
second_moment_offdiag_12=+0.0569; second_moment_offdiag_13=+0.473
second_moment_offdiag_23=-0.59; cog_x=+0.0231
cog_y=-0.0575; cog_z=+0.0152
 $\log_{\text{force_eff_1}} = +0.223$; $\log_{\text{force_eff_2}} = +0.47$
 $\log_{\text{force_eff_3}} = -0.131$; $\log_{\text{force_eff_4}} = -0.282$

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

force_over_mass_rotor_3_nominal: optimization vs reference covariance



force_over_mass_rotor_3_nominal: wrench / closure (force max 4.86e-15, torque max 4.44e-16)



force_over_mass_rotor_3_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_force_over_mass_rotor_3_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/force_over_mass

SHA256: 6b7aab340fa6fd878752d7c8d399d9306f6b9fcfbe1938694085b3ade666560d

data objective: 37.823430149

prior objective: 6.50783906425e-08

total objective: 37.8234302141

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: force_over_mass_rotor_3_nominal (force_effectiveness_over_mass)

components: rotor_3

target: [0.42524282]

std: [1.e-05]

final: [0.42524282]

error: [-3.60772479e-09]

standardized residual: [-0.00036077]

factor objective: 6.50783906425e-08